

PROJECT AND TEAM INFORMATION

Project Title

(Try to choose a catchy title. Max 20 words).

FoodRescue — Smart Food Rescue and Redistribution System

Student / Team Information

Team Name: Team #	Avengers
Team member 1 (Team Lead) (Last Name, name: student ID: email, picture):	Dutta, Aditya – 2510373905 2510373905@geu.ac.in 
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PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

(Describe the problem you want to solve and why it is important. Max 300 words).

Restaurants, hostels, households, and events often generate surplus food that is still safe to consume, while NGOs and people in need may face shortages. Without proper coordination, usable food is wasted. FoodRescue connects donors with NGOs by recording food quantity, location, type, and expiry time. It prioritizes donations using expiry, quantity, and distance to support efficient redistribution. The project demonstrates how DSA and OOP can solve a practical food-wastage and resource redistribution problem.

State of the Art / Current solution (1 pt)

(Describe how the problem is solved today (if it is). Max 200 words).

Surplus-food redistribution is currently handled through NGOs, volunteers, phone calls, messaging apps, social media groups, and donation platforms. These methods work, but coordination becomes difficult with many donors, NGOs, volunteers, locations, and time-sensitive food. Manual coordination also struggles to consider expiry time, quantity, and distance together. FoodRescue addresses this using priority queues, queues, hash tables, graphs, vectors, and matching algorithms for efficient redistribution.

Project Goals and Milestones (2 pts)

(Describe the project general goals. Include initial milestones as well any other milestones. Max 300 words).

The main goal is to develop a C++ system using OOP and DSA to connect surplus-food donors with NGOs and manage redistribution efficiently. Key milestones are:

- 1. Understand the requirements and create the main system entities and classes.*
- 2. Manage donors, NGO, food, requirement, and volunteer management.*
- 3. Select food based on priority and manage donors with suitable NGO's.*
- 4. Design a graph to find best route using Dijkstra's algorithm.*
- 5. Add delivery tracking, test the system, and show how food moves from donor to people in need.*

Project Approach (3 pts)

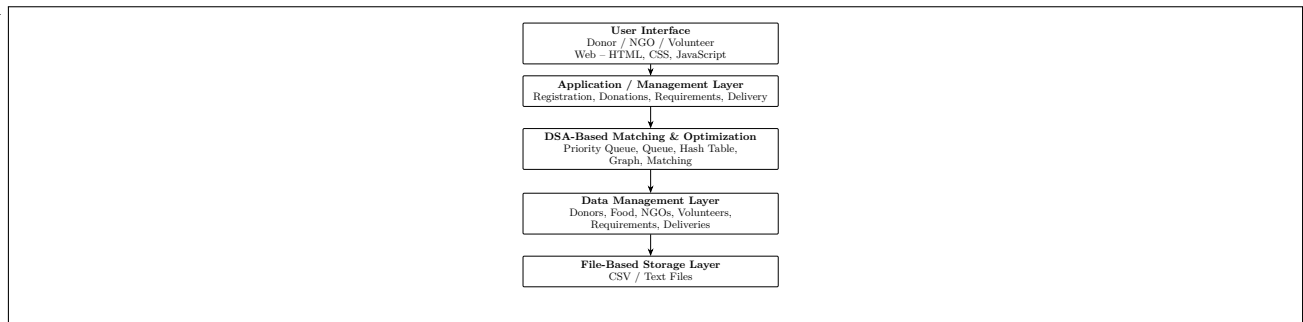
(Describe how you plan to articulate and design a solution. Including platforms and technologies that you will use. Max 300 words).

FoodRescue will be developed in C++ using OOP and DSA along with HTML, CSS and JavaScript. The system will have classes like Donor, FoodDonation, NGO, Volunteer, Requirement and Delivery. Priority queues will be used to manage urgent food donations, queues will handle requests, and hash tables helps find data quickly. Graphs will be used to represent locations and roads. Food donations will be matched based on expiry time, quantity and distance. Dijkstra's algorithm will be used to find suitable delivery routes. The system will also track each donation from registration until delivered.

System Architecture (High Level Diagram)(2 pts)

(Provide an overview of the system, identifying its main components and interfaces in the form of a diagram using a tool of your choice).

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Project Outcome / Deliverables (1 pts)

(Describe what are the outcomes / deliverables of the project. Max 200 words).

FoodRescue is a C++ based system that connects surplus-food donors with NGOs and manages the redistribution process. The system will provide donor, NGO, food, requirement, volunteer and delivery management. It will include priority-based donation selection, donor-NGO matching, graph-based route planning using Dijkstra's algorithm, delivery tracking and record storage along with a web-based front-end (HTML, CSS, JavaScript) for donor, NGO and volunteer interaction. Additional deliverables include the complete C++ source code, documentation and a final working demonstration.

Assumptions

(Describe the assumptions (if any) you are making to solve the problem. Max 100 words)

The system assumes that donor, food, NGO, volunteer, and location information is reasonably accurate. Volunteers are assumed to have transportation for assigned deliveries, and road networks are represented using predefined nodes and edges.

Food is assumed to be safe for consumption when registered. Food safety certification, live GPS tracking, and real-time map integration are outside the initial scope of the system. The system will use stored datasets and file-based records.

References

(Provide a list of resources or references you utilised for the completion of this deliverable. You may provide links).

1. Cormen, T. H., Leiserson, C. E., Rivest, R. L., and Stein, C. — Introduction to Algorithms.
2. Balagurusamy, E. — Object Oriented Programming with C++.
3. C++ Standard Template Library (STL) documentation.
4. Course material and lecture notes on Data Structures and Object-Oriented Programming.